

Adaptive Experience Architecture

Adaptive Experience = Shared Understanding + Domain Services + Outer Harness

The 6-Layer Production Playbook & The 5-Floor Concentric Harness Hierarchy

Instantiated on Lily's Florist Path B (Art of Group). Synthesizing the harness taxonomy of Hashimoto, Fowler, Codex, 0xWast3 (Memory Engineering), and Kocer (Five Concentric Floors). Canonical remote: [GitLab gitlab.com/artof-group/adaptive-experience-architecture](https://gitlab.gitlab.com/artof-group/adaptive-experience-architecture). Live shop: aea.artof.link. Architecture site: architecture.artof.link. Tracker is [GitLab \(glab\)](https://gitlab.com), not [GitHub](https://github.com).

Independently compiled, Revision of 14 September 2026 (agent-OS Keep Learning addendum on 10 Sep / 31 Aug base) — Art of Group / AEA knowledge guardian — not affiliated with Google, OpenAI, Anthropic, HashiCorp, or Moonshot AI — and not endorsed. Template inspired by [harness_final.pdf](#).

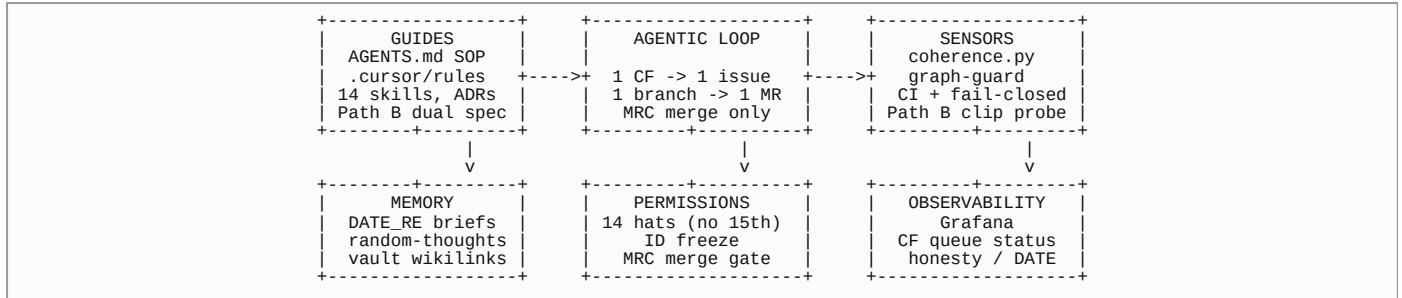


Fig. 1. The six-layer Adaptive Experience Architecture mapped onto the outer harness. Guides and sensors form the steering loop (feedforward + feedback). Memory, permissions, and observability form the runtime foundation. The agentic loop sits at the center: one finding, one GitLab issue, one branch, one MR.

Abstract — Production AI-native experiences fail the same way production autonomous agents fail: a fluent demo is not a bounded, honest, merge-gated system. This paper presents Adaptive Experience Architecture (AEA) as an outer harness around an enterprise florist product, Lily's Florist Path B, where customers speak naturally and the interface evolves as shared understanding grows. The core formula is Adaptive Experience = Shared Understanding + Domain Services + Outer Harness. The model is whatever LLM Path B calls through LiteLLM under the NFR-003 timeout SLA ($\leq 2.5s$). The outer harness is the 14-role stakeholder team, the Second Brain vault, the coherence finding loop, and CI pre-flight quality guards. The inner product runtime is the edge BFF, platform domain services, PostgreSQL 16, and an event-driven broker per the ADRs. We map AEA onto six outer harness layers and five concentric wrapping floors (Prompt, Context, Harness, Loop, Graph), reporting only findings that have been probed. Related work is cited, not inherited as evidence.

Index Terms — Adaptive Experience Architecture, outer harness, shared understanding, Path B dual viewport, coherence findings, DATE_RE, 14 hats, fail-closed inventory, GitLab MR coordinator, honesty rule, context curator, orthogonal reviewer.

I. THE PROBLEM: DEMOS THAT NEVER SHIP

Why a fluent florist demo is not yet a production experience

A chatbot produces conversational responses. An AI-native florist must produce deterministic outcomes: a validated recommendation, an honest availability badge, an itemized order summary, and a session that survives a viewport transition. The difference is not which model Path B called this week. The difference is the engineering infrastructure that turns a probabilistic interpreter into a bounded, observable shop. Path B lets customers speak naturally. The interface is supposed to evolve as understanding grows. That promise dies the moment the system performs confidence it did not earn.

AEA already has an exact name for that failure class: **CF-048** is the daily-brief honesty finding. Status words appeared without a probe. The DATE_RE generator can claim coverage it did not measure. A session can narrate a ship that the vault does not show. That is the AEA version of demos that never ship: not a missing model, a missing harness. CF-048 was verified through GitLab issue #259 and MR !280. The lesson is the product: fluent language around an unprobed status is a regression, not a milestone.

A. The Three Eras, for Experience Systems

The industry spent 2023–2024 optimizing prompts. It spent 2025 stuffing context: RAG, MCP, memory. By 2026, coding-agent practitioners named a third discipline: harness engineering [1], [2], [3]. Experience systems need the same progression, with different artifacts. A prompt does not keep FR-011 honest. A larger context window does not stop a knowledge MR from closing a UI finding. The outer harness does.

TABLE I — THREE ERAS OF EXPERIENCE ENGINEERING

ERA	FOCUS	OPTIMIZES	LIMITATION
Prompt Eng. (2023–24)	Single turn	Phrasing, tone	One utterance
Context Eng. (2025)	What model sees	RAG, MCP, SU	Info, not action
Harness Eng. (2026)	Full environment	Guides, sensors, merge	Entire runtime + shop

Each era subsumes the previous one. AEA Path B still needs prompts (the concierge voice) and context (Shared Understanding, FR-001/021). The harness is what decides whether those turns become a shop a customer can trust. Prompt engineering shapes what the model says. Context engineering shapes what the model sees. Harness engineering shapes what the experience is allowed to claim, what domain services must validate, and what constitutes a finished journey [3].

B. Who This Playbook Is For

This paper is for the 14 AEA stakeholder hats, and for product teams building an AI-native consumer experience rather than a standalone coding agent. No new agent framework is required. The architecture is useful wherever an LLM interprets intent and domain services must remain the authoritative source of truth. 3DX Lab is a separate project and is out of scope here. Tracker is [GitLab \(glab\)](https://gitlab.com); [GitHub](https://github.com) is a one-way mirror.

II. THE FORMULA: ADAPTIVE EXPERIENCE = SU + SERVICES + HARNESS

How Path B restates Agent = Model + Harness

Related work crystallized a coding-agent formula: Agent = Model + Harness [1], [2], [3]. Hashimoto's ratchet is the method: when the agent makes a mistake, engineer a solution so the mistake does not recur [1]. AEA keeps that ratchet and changes the product. The thing being shipped is not a million-line coding agent. It is a florist experience in which Shared Understanding is editable, recommendations are validated, and availability is fail-closed.

"Anytime you find an agent makes a mistake, you take the time to engineer a solution such that the agent never makes that mistake again."

— Mitchell Hashimoto [1]

The AEA restatement is operational. Shared Understanding is the customer-visible state (FR-001, FR-021). Domain services validate what the model is not allowed to invent (FR-007, inventory, delivery, payment). The outer harness is everything that keeps fourteen hats from improvising a fifteenth, keeps IDs frozen, and keeps merges behind [@aea-mr-coordinator](#).

TABLE II — INNER RUNTIME VERSUS OUTER HARNESS (AEA)

LAYER	WHAT IT IS	WHAT IT IS NOT
Model	LLM Path B calls via LiteLLM; NFR-003 ≤ 2.5s	The product
Inner runtime	Edge BFF, domain services, PostgreSQL, Kafka broker	The 14-hat team
Outer harness	14 roles, vault, CF loop, CI guards, DATE_RE	A prompt file
Shared Understanding	Editable customer state (FR-001/021)	A hidden chain-of-thought

A. Related Work, Not Inherited Evidence

Third-party harness write-ups report impressive gains: GAIA swings, Terminal Bench ranks, million-line codebases [2], [6], [7]. Those numbers are not AEA results. AEA evidence is narrower: CF-048 verified, dual-viewport trail on !298 / !299 / !300, fail-closed inventory (FR-011 / NFR-009), and the honest leftover list from 2026-08-27 clips.

B. Inner Harness vs. Outer Harness

Frontier labs ship an inner harness: safety layers, native tool calling, context windows. AEA does not replace that. The engineering moat for Path B is the outer harness: session-start SOP, 14 canonical skills, coherence scripts, GitLab gates, fail-closed inventory, and the honesty rule [5]. Domain services validate. The broker stays product-neutral per the ADRs. Payment FR-019 is mockup, not live Stripe. That is an honesty constraint, not a roadmap slogan.

III. LAYER 1: GUIDES

Feedforward controls that prevent known failures before a session starts

Guides are what an AEA session reads before it acts. They are feedforward. Fowler and Böckeler's guides-and-sensors taxonomy remains the vocabulary [3]. On Path B the guide surface is specific, not generic folklore.

A. What Guides Contain

The session-start SOP lives in AGENTS.md. Cursor loads .cursor/rules with session-start-briefing, coherence-findings-sop, docker-integration-before-mr, and stakeholder-skills-sync. Fourteen canonical skills sit under .cursor/skills/aea-*/. ADRs freeze architecture moves. FR and NFR IDs come from the workbook. Path B CSS is owned by @aea-ux-designer. Motivational language is not a guide.

```

Template 1 – Minimum AEA session-start guide
PROJECT: Lily's Florist / Path B
TRACKER: GitLab (glab). GitHub is a one-way mirror.
HANDOFF: research/daily-briefs/YYYY-MM-DD.md # DATE_RE only
RULES:
- Do not invent BG / US / FR / NFR IDs
- Do not invent a 15th implementer hat
- One CF -> one issue -> one branch -> one MR
- Only @aea-mr-coordinator merges
ANTI-PATTERNS:
- Status words without a probe (CF-048)
- Loop ticks that merge
- Knowledge MR closing a UI finding

```

Vision principles from docs/01-product-vision/product-vision.md belong in the guide layer because they constrain design: Thought before form; Knowledge before navigation; Shared understanding before recommendation; Experiences earn attention; Continuity before immediacy; Latest relevant intent wins; AI interprets, domain services validate. BG-001 through BG-007 exist in that vision. This paper does not invent their wording.

B. The Ratchet, AEA-shaped

Hashimoto's ratchet still holds [1]. The six steps on Path B: (1) A session or clip surfaces a failure. (2) Classify the class, not the anecdote. (3) Pick the strongest layer: guide, sensor, permission, or merge policy. (4) Encode it in the vault. (5) Open one CF and one issue. (6) Do not call it verified until a probe says so. A prompt patch fixes one conversation. A skill rule fixes every future hat that loads it. A CI guard makes the error structurally harder.

C. Guide Hygiene

Fourteen skills and four always-on rules already accumulate. Without hygiene they contradict. Version them. Date the reason a rule exists. Remove a rule when a sensor now enforces it. Do not grow a fifteenth skill that is secretly "just implement it." A guide file with undated aspirations is not a harness. It is vault drift.

- Can the hat verify the rule without subjective taste?
- Does the rule trace to an observed CF, ADR, FR, or NFR?
- Are two skills telling the same hat opposite merge policies?
- Could a python guard replace this sentence?
- When was the rule last read against current Path B behavior?

D. Guides as Organizational Memory

AGENTS.md is not a suggestion to the model. It is the system of record for how a session must begin. When the conversation and the guide conflict, the guide wins. That inversion is what makes the outer harness durable [2]. New hats inherit the ID freeze and the MRC merge rule in seconds. When a human leaves, the corrections remain in skills, ADRs, and the CF queue.

E. The Cost of Not Having Guides

Without the SOP, every new session infers Path B from the live shop alone. It will invent FR IDs. It will treat GitHub as the tracker. It will mark CF-054 verified because CSS merged. It will write a DATE_RE file that is not the DATE_RE filename. Every correction made only in chat dies with the session. The guide file is what makes the correction permanent.

IV. LAYER 2: SENSORS

Feedback controls; computational first; clips are probes, not trophies

Sensors verify after execution. They catch what guides did not prevent. Computational sensors first [3]. Inferential judgment is expensive and non-deterministic. Path B already has python guards. Use them before asking a model whether the shop "feels" done.

TABLE III — AEA SENSOR TYPES AND RELIABILITY

TYPE	EXAMPLE	SPEED	DETERMINISM
Computational	scripts/check_coherence.py	Fast	Deterministic
Computational	check_knowledge_graph.py	Fast	Deterministic
Computational	run_all_guards.py + GitLab CI	Fast	Deterministic
Computational	Fail-closed inventory FR-011/NFR-009	Fast	Deterministic
Probe (UI)	Path B journey×viewport clips (CF-054)	Slow	Human-scored
Inferential	LLM-as-judge on copy tone	Slow	Non-deterministic

Fail-closed inventory is a product sensor, not a linter. Missing or stale `observed_at` becomes `unknown`. Select is disabled. That is FR-011 / NFR-009. A green Available badge without a fresh observation is a demo. DATE_RE claiming 14/14 skill coverage is a generator claim. This paper does not restate 14/14 as independently probed without verification.

A. Self-Verification Pattern

The strongest pattern gives the session its own sensors. After a change, run the guard. Loop the error text back. This is not the model grading itself. It is the model executing an external check.

```

def fail_closed_inventory(row):
    if missing(row.observed_at) or stale(row.observed_at):
        return "unknown" # Select disabled (FR-011 / NFR-009)
    return row.availability

def verified_step(hat, task, sensors):
    result = hat.execute(task)
    for sensor in sensors:
        verdict = sensor.check(result)
        if not verdict.passed:
            result = hat.fix(result, verdict)
        if not sensor.check(result).passed:
            return escalate(task, verdict)
    return result

```

B. Clip Probes

Path B journey×viewport clips are UI probes (CF-054), not marketing. They sit below computational sensors because they are slow and human-scored. They still matter: a dual-viewport shop that only works in one recording is not dual. J1 (Urgent Sam), J2 (Planner Sarah), J3 (Loyal Alex), and J4 (Tracker Chris) are the journey set. One session, desktop Adaptive Workspace, phone linear concierge. Honest leftovers from the 2026-08-27 clips: J3 recall without a reorder badge; J4 ASO fail-closed, not T-08; Track gated until checkout. Live shop re-record of J1 phone+desktop after the 2026-08-28 CSS merge is Unknown. Do not mark CF-054 verified.

C. Sensor Economics

`check_coherence.py` costs nothing per run compared with a token-heavy judge. `graph-guard` catches ID drift before a hat invents US-008. CI is the sensor that survives a closed laptop. Add inferential

sensors only for checks no deterministic rule can capture, such as concierge tone. Treat those as advisory, not gates.

D. The Sensor Coverage Test

- Can a session check inventory honesty without a human staring at Grafana?
- Does every critical Path B claim have at least one computational sensor?
- Are sensor results logged so CF status is not a vibe?
- Could a silent DATE_RE lie reach the next session without tripping a wire?
- If any answer is no, add the sensor before claiming unattended operation.

V. LAYER 3: THE AGENTIC LOOP

One finding, one issue, one branch, one MR; loop ticks must not merge

The AEA agentic loop is not Plan-Execute-Retry inside one context window. It is a typed GitLab loop. The CF-NNN queue lives in `research/coherence-findings-loop.md`. The unit of work is one finding. Fan-out is how honest queues die.

```
def cf_loop(finding, vault):
    issue = gitlab.open_issue(finding) # one CF -> one issue
    branch = spawn_branch(issue) # one issue -> one branch
    mr = open_mr(branch) # one branch -> one MR
    if loop_tick(mr):
        return reject("loop ticks must not merge")
    if not gates_pass(mr):
        return escalate(issue, packet) # PO / PM / sponsor
    return mrc.merge(mr) # only @aea-mr-coordinator
```

TABLE IV — REQUIRED LOOP BOUNDS (AEA)

BOUND	PURPOSE	DEFAULT
One CF per issue	Prevent queue fan-out	Hard rule
Loop ticks merge?	Protect main	Never
Who merges	Independent verifier	@aea-mr-coordinator only
Auto-merge	When gates pass	MRC policy, not producing hat
Escalation	Human decision packet	PO / PM / sponsor
Stopping condition	Honest partial state	Always defined

A. Escalation Is Not Failure

A hat that escalates with a packet is more valuable than one that merges a fluent wrong shop. The packet should contain: the decision required, the recommended option, alternatives already tested, cost of waiting, and the safest default if nobody replies. Secrets and budget escalate to the sponsor. Archive and workbook edits need human confirmation. Path B CSS changes escalate to @aea-ux-designer, not to whoever has the editor open.

B. The Background Agent Rule, Restated

Hashimoto wants an agent working while the human reviews [1]. AEA allows background work inside a finding. It does not allow background merges. Humans steer hats. MRC steers main. That split is the loop.

VI. LAYER 4: MEMORY AND STATE

The model forgets every session. Context is a workspace, not memory

Every model call starts empty. A 1M-token context window is a temporary workspace, not durable memory [19]. Replaying full history at session start is expensive and silent: a throwaway comment weighs the same as a hard constraint. The harness reconstructs state across distinct memory primitives:

TABLE V — STATE PERSISTENCE LAYERS (AEA)

LAYER	PERSISTS	IMPLEMENTATION
Context window	Current turn buffer	In-memory conversation buffer
Procedure memory	Repeatable workflows	14 Skills (.cursor/skills/aea-*/)
Correction memory	Active negative rules	AGENTS.md & Second Brain
Relationship memory	Typed graph links	Obsidian [[wikilinks]] + #aea
Handoff brief	Cross-session state	research/daily-briefs/YYYY-MM-DD.md
Git working copy	Committed main	Only source of shared truth

A. DATE_RE versus random-thoughts

`random-thoughts/` is allowed to be messy. DATE_RE is not. Cadence jobs must not write DATE_RE. A generator that pollutes the

handoff filename recreates CF-048. The brief is a state file. Treat it like a checkpoint, not a blog.

```
BRIEF = f"research/daily-briefs/{today}.md" # DATE_RE only
# recovery test: new session reads the committed brief
# uncommitted files are not shared memory
# cadence must not write DATE_RE
```

B. The Recovery Test

Close the session in the middle of a finding. Open a new one. It must read the committed DATE_RE brief, identify the open CF, and resume without asking a human to re-explain Path B. If it cannot, memory is insufficient. Do not compensate by pasting yesterday's chat. That is not a harness.

C. Memory versus Guides

Memory records what happened. Guides define what should happen. Promote a stable policy into a skill or a rule. Leave the day's debris in `random-thoughts/`. Do not rely on an agent remembering the ID freeze from a prior conversation.

VII. LAYER 5: PERMISSIONS AND BUDGETS

Fourteen hats. No fifteenth implementer. The harness is the security boundary

The model cannot restrict itself. Permissions are a harness property [3]. AEA encodes them as roles, ID freeze, merge policy, and human confirmation gates.

```
Template 2 — AEA capability budget
ALLOW: role-scoped skill under .cursor/skills/aea-*/
ALLOW: one CF -> one issue -> one branch -> one MR
DENY: invent BG / US / FR / NFR IDs
DENY: invent a 15th implementer hat
DENY: knowledge MR closing a UI finding
ASK: archive/ or workbook changes (human confirmation)
HUMAN: secrets, budget (sponsor)
CSS Path B: @aea-ux-designer
MERGE: @aea-mr-coordinator only; no casual merge
```

TABLE VI — DEFAULT ACTION POLICY (14 HATS)

ACTION	DEFAULT	REASON
Read vault + skills	Allow	Reversible observation
Open CF + issue	Allow	Typed intake
Edit Path B CSS	@aea-ux-designer	Dual-viewport ownership
Change workbook IDs	Human confirm	ID freeze
Merge to main	MRC only	Independent verifier
Secrets / spend	Sponsor	Irreversible
Invent a hat or ID	Deny	Blast radius

The fourteen hats are: *aea-project-manager*, *aea-product-owner*, *aea-ux-designer*, *aea-customer-journey*, *aea-support-coordinator*, *aea-ai-engineer*, *aea-appsec-auditor*, *aea-devsecops-platform*, *aea-senior-software-engineer*, *aea-mr-coordinator*, *aea-coherence-guardian*, *aea-knowledge-guardian*, *aea-cost-guardian*, *aea-performance-guardian*. Typed handoffs travel through GitLab issues/MRs and the vault. MRC is the independent verifier. Producer bias is why the producing hat does not merge.

A. Untrusted Surfaces

Path B reads customer speech. That is untrusted content. A crafted utterance must not expand permissions, mint an FR ID, or trigger a merge. Trusted instructions are guide files and skills. Untrusted data is customer input, retrieved docs, and clip narration. Keep them split [2].

B. Four Budget Dimensions

Scope: which hat, which files, which GitLab actions. Rate: findings per session, not an unbounded rewrite of the shop. Reversibility: workbook edits and live CSS are harder to reverse than a draft MR. Visibility: CF status, Grafana, and the brief must show who did what. Encode all four before unattended runs. NFR-003 ($\leq 2.5s$) is a latency budget on the model call, not permission to skip sensors.

VIII. LAYER 6: OBSERVABILITY

Status words need a probe. Grafana is not a vibe check

A production harness needs telemetry. Required on Path B: CF queue Last seen / status, guard verdicts, LiteLLM latency via CloudWatch, and Grafana at <https://aea.artof.link/grafana/>. The honesty rule is the distinctive AEA sensor: status words need a probe. Without it, a faster concierge looks successful while omitting the claim that mattered.

TABLE VII — AEA TRIP WIRE TRIGGERS

TRIP WIRE	INDICATES	RESPONSE
DATE_RE honesty (CF-048)	Unprobed status words	Reject the brief
graph-guard fail	ID or link drift	Block the MR
Cadence writes DATE_RE	Handoff pollution	Stop the job
Stale observed_at	Inventory demo-mode	unknown + Select disabled
CF row vs merged CSS	Queue lag (CF-054 class)	Do not claim verified
NFR-003 breach	LiteLLM SLA miss	Inspect CloudWatch

TABLE VIII — HARNESS HEALTH (NO FAKE TARGETS)

SIGNAL	DEFINITION	DIRECTION
Verified CFs	Rows with an actual probe	Up, honestly
Unprobed status words	Claims without evidence	Down to zero
Queue lag	Merged work still in-mr on main	Down
Fail-closed trips	unknown shown when stale	Must fire
MRC merges	Non-MRC merges	Down to zero
DATE_RE pollution	Non-handoff writers	Down to zero

A. The Real Metric

Do not count tokens, messages, or hats invoked. Count findings that closed with evidence and left the shop honest. This paper states no completion-rate percentage. That number was not probed here. Unknown is the correct word when the probe is missing.

B. Observability as Debugging Infrastructure

When Path B is wrong, the first question is where. Grafana, CloudWatch, the CF row, and the MR pipeline should answer without replaying the whole conversation. Observability is not overhead. It is how CF-048 was catchable.

C. Cost Attribution

Cost-guardian exists as a hat. This compilation does not publish a dollars-per-journey figure. Unknown. Track cost per verified finding when the probe exists, not per chatty session. LiteLLM spend without an honest shop is not efficiency.

Operational Review Questions:

- Can you name the CF that last changed status, and the probe that justified it?
- Can you see LiteLLM latency against NFR-003 in CloudWatch?
- Can you tell DATE_RE from a random-thoughts log in one glance?
- Would you know if cadence wrote the handoff file?
- Would you know if main still says in-mr after CSS merged?

IX. THE CONCENTRIC FIVE-FLOOR WRAPPING MODEL

Harness, Loop, and Graph are nested floors, not competing choices

In August 2026, Kocer resolved the industry debate between harness, loop, and graph architectures [21]. They are not competing choices; they are **five concentric wrapping floors** where each floor assumes the stability of the floor beneath it:

01 | Prompt Engineering (The Message): Unit of work: One input (role, task instructions, examples, formatting constraints → single raw response).

02 | Context Engineering (The Memory): Unit of work: What stays in the window. A curator selects, compresses, and drops from query, docs, memory, prior turns, and tool outputs before the prompt runs.

03 | Harness Engineering (The Machine): Unit of work: The execution machine itself. Gather (context + prompt) → LLM → tools → verifier → final response. The operating environment.

04 | Loop Engineering (The System): Unit of work: The run. Goal + success criteria + max iterations + budget + completion check wrapping one harness pass. Failed pass appends results to context and retries.

05 | Graph Engineering (The Topology): Unit of work: The graph run. Goal + nodes + edges + state schema. Routes to agent nodes, tool nodes, or human approval. An independent reviewer node with a different model and fresh context checks the final answer.

A. Three Invariant Laws of Concentric Wrapping

1. The Lower-Floor Dependency Law: Layer 5 assumes Layer 4 works; Layer 4 assumes Layer 3 works; Layer 3 assumes Layer 2 works.

If Layer 2 context curation is skipped, Layer 3 verifiers fail cryptically, and Layer 5 graphs look broken when standing on nothing.

2. The Orthogonal Reviewer Law: In Layer 05, verification must be executed by an independent reviewer node operating with a different model family in an unpolluted, fresh context window.

3. The Economic Moat Law: Swapping the foundation model is a 1-day commodity task (via LiteLLM under ADR-016). Swapping the 5-layer engineering harness takes a quarter. The model is the commodity; the five layers around it are the proprietary engineering asset.

X. THE IMPROVEMENT LOOP & RELIABILITY LADDER

CF intake is the ratchet. Clip re-record is a sensor, not a launch party

TABLE IX — FAILURE CLASSIFICATION (AEA)

FAILURE CLASS	WEAK FIX (AVOID)	STRONG FIX (PREFER)
Unprobed status (CF-048)	Rewrite sentence	Honesty rule + DATE_RE trip
Invented FR/NFR ID	Correct in chat	graph-guard + ID freeze
Knowledge closes UI CF	Comment on MR	Permission + MRC policy
Stale availability	Softer copy	Fail-closed Select disable
Dual-viewport drift	Looks fine on laptop	Spec + UX CSS + clip probe
Loop tick merge	Revert later	MRC-only merge gate
Lost session context	Re-explain Path B	Committed DATE_RE brief

"Prompts guide behavior. Environments prevent entire classes of failure."

— Lauren Tan, Cursor [10]

CF intake is the ratchet. Each failure that lands as a numbered finding, with one issue and one MR, reduces the chance the next session repeats it. Early in a vault, guide growth is fast. A mature harness adds fewer rules and more guards. This paper does not claim a rules-per-week rate. Unknown.

A. The Six-Step Engineering Loop

When Path B fails: (1) Reproduce with the same journey and viewport. (2) Classify: missing guide, absent sensor, permission gap, state loss, or observability blind spot. (3) Pick the strongest layer. (4) Encode the fix. (5) Verify on the original case. (6) Run guards so you did not break FR-011 while fixing copy. Skip no step. Skipping (1) fixes a phantom. Skipping (5) is how CF-054 stays in-mr on main after the CSS has merged.

B. Clip Re-record as Ratchet

Re-record J1–J4 when the dual-viewport spec or CSS changes. A clip that predates !300 is not evidence that !300 works on the live shop. The 2026-08-28 trail is: knowledge !298 merged, spec !299 merged, feat(ux) CSS !300 merged, issue #272 closed. The CF-054 row on main still said in-mr pointing at !298. Live J1 phone+desktop re-record: Unknown.

TABLE X — CONTROL RELIABILITY LADDER

LAYER	AEA EXAMPLE	RELIABILITY
Memory	Correction in chat	Low
Prompt	Be honest in the brief	Low-Medium
Guide	AGENTS.md + skill rule	Medium
Sensor	check_coherence.py / fail-closed	High
Environment	MRC merge + CI + ID freeze	Highest

XI. BUILD PATH: HOW AEA ALREADY INSTANTIATED THE LAYERS

Not a fake seven-day greenfield. The vault is the path

Do not pretend AEA was stood up in a week on an empty repo. The six layers are already instantiated, unevenly. The build path is to notice which layer is thin and strengthen it from real CFs. Roadmap group milestones are M0–M7 MVP, M8–M12 post-MVP, M19 native mobile companion, then Future Backlog. DATE_RE 2026-08-28 listed Active Focus Unknown. Milestone lines are roadmap labels, not ship-counts. Do not claim M14–M18 production-ready.

TABLE XI — AEA LAYER INSTANTIATION (AUGUST 2026)

LAYER	ALREADY IN VAULT	HONEST GAP
Guides	AGENTS.md, 4 rules, 14 skills, ADRs, Path B spec	Hygiene vs sprawl
Sensors	coherence / graph / guards / CI / fail-closed	DATE_RE 14/14 unprobed here
Loop	CF queue, one-finding rule, MRC merge	Queue lag (CF-054 row)
Memory	DATE_RE filename, random-thoughts, wikilinks	Uncommitted-as-memory temptation
Permissions	14 hats, ID freeze, UX CSS ownership	15th-hat gravity
Observability	Grafana, CF Last seen, CloudWatch, honesty	Clip re-record Unknown

A. What MVP Actually Includes

Requirements count is 23 FR + 17 NFR = 40. Source of truth: [archive/Quantic_Project_Consolidated_Coherence_Validated.xlsx](#). Cite existing IDs only. MVP includes conversational discovery, editable Shared Understanding (FR-001/021), validated recommendations (FR-007), T-03 Available badge (FR-011/NFR-009), thin T-04 FR-003 (size, card message, flower_type/colour/ribbon per ADR-006), delivery, itemized summary FR-018, payment FR-019 (mockup / not live Stripe), tracking FR-023, FAQ FR-009.

B. What Is Not Shipped

Future / not shipped: live Stripe, browser LCP / Edge SSR, staff live chat, WebRTC, full CRM analytics FR-017, inventory forecasting FR-012. Thin CRM occasion reminders landed as #254/!297 and remain FR-016 Future in the workbook. Naming a merge is not the same as moving the workbook ID.

C. Expansion Pattern

After the layers exist, expansion follows the ratchet. Run Path B. Every failure names a layer. Same error twice: guide hole. Subtly wrong shop: sensor hole. Scope creep: permission too broad. Cannot diagnose: observability hole. Change one layer at a time. Do not scale hats because the model is impressive.

Scale Gate — Expand Only When These Are True:

- MRC, not the producer, is the only merger on main.
- Fail-closed inventory has tripped on a stale observation, not only on happy path.
- DATE_RE recovery test passed on a new session.
- Honesty rule has caught at least one unprobed status word.
- A clip probe exists for the journey you claim, dated after the CSS that claimed it.
- No fifteenth hat has been introduced as just this once.

XII. DECISION FRAMEWORK

Start at the simplest layer that matches the observed failure

TABLE XII — ARCHITECTURE DECISION FRAMEWORK (AEA)

PROBLEM	START WITH	DO NOT ADD YET
Known errors repeat	Guide / skill rule	Another hat
Availability lies	Fail-closed sensor	Softer copy
Hat exceeds scope	Permission + ID freeze	Full committee every edit
State lost next session	Committed DATE_RE	Vector memory
Status words drift	Honesty trip wire	LLM-as-judge on the brief
Viewport divergence	Spec + UX-owned CSS	A second product
Loop never stops	One-CF + MRC	Multi-agent debate

Run the experience. Observe failures. Let the harness grow from CFs. Do not build a heavier harness than the shop. Do not skip sensors because the concierge sounded sure.

XIII. PRODUCTION CHECKLIST FOR PATH B + ENGINEERING HARNESS

A shop is not production because the demo clip is pretty

TABLE XIII — PATH B PRODUCTION READINESS CHECKLIST

#	REQUIREMENT	FAILURE IF MISSING
1	AGENTS.md SOP + DATE_RE handoff	Every session starts from zero
2	14 hats, no 15th	Shadow implementer
3	FR/NFR ID freeze + graph-guard	Workbook drift
4	Computational guards in CI	Errors reach the shop
5	Fail-closed FR-011 / NFR-009	Available on stale data
6	One CF / issue / branch / MR	Queue fan-out
7	MRC-only merge	Producer bias on main
8	Honesty rule (CF-048 class)	Unprobed status words
9	Grafana + CloudWatch NFR-003	Blind latency and drift
10	Trusted/untrusted split	Speech as instruction
11	UX ownership of Path B CSS	Viewport fork
12	Clip probe dated after the CSS	Claim without evidence

XIV. WHEN NOT TO GROW THE HARNESS

Not every utterance justifies another skill

A harness earns its cost when the experience repeats, failures have customer consequences, sessions must preserve Shared Understanding, or more than one hat touches main. One-off copy brainstorm, exploratory product talks, and throwaway prototypes do not need DATE_RE, MRC, and fourteen skills. Path B is not that. It is external-facing and must stay honest about stock, payment mockups, and tracking.

TABLE XVI — HARNESS DECISION FILTER

SCENARIO	HARNESS?	WHY
One-off copy riff	No	No recurrence
Exploratory UX sketch	Maybe	If it will hit Path B CSS, yes
Live availability claim	Yes	FR-011 customer consequence
Daily DATE_RE handoff	Yes	Next session depends on it
CF intake / MRC merge	Yes	Main is shared
Personal note in random-thoughts	No	Not shared memory
Payment FR-019 mockup	Yes	Must not imply live Stripe
3DX Lab work	Separate	Out of scope for this paper

The test: would a customer notice a silent wrong Available badge? Then you need a sensor. Would the next session need yesterday's intent? Then you need DATE_RE. Would a mistaken merge change the shop? Then you need MRC. If none apply, the conversation itself is enough.

XV. HARNESS ENGINEERING FOR THE FOURTEEN HATS

Multi-agent here means typed hats, not a swarm

The six-layer harness applies to each hat. The system additionally needs typed handoffs, a shared vault, a routing policy (the CF queue), and an independent verifier no producer can override. That verifier is @aea-mr-coordinator.

A. Typed Handoffs

When @aea-ux-designer passes work to @aea-senior-software-engineer, the handoff is a GitLab artifact: issue, MR, evidence, unresolved remainder. "Done, looks good" must not advance Path B. The receiver should open the spec or the CSS and check. Packet fields still apply: identity, pointer, evidence, assumptions, deadline [11].

B. Shared Memory versus Shared Context

Fourteen hats in one conversation pollute context. AEA's better pattern is already in the vault: hats read DATE_RE, the CF row, and the skill that matches the hat. They do not inherit another hat's discarded plan. Uncommitted files remain private.

C. The Verifier Must Be Independent

The hat that produced the CSS is not the judge of whether CF-054 is verified. Producer bias is real. Deterministic guards first. MRC reports gate failures without silently rewriting the shop. Auto-merge when gates pass is still an MRC act, not a courtesy from the producer.

TABLE XVII — FOURTEEN-HAT HARNESS EXTENSIONS

MULTI-HAT LAYER	SINGLE-HAT EQUIVALENT	WHAT IT ADDS
GitLab issue / MR packet	Self-check	Contract between hats
Vault + DATE_RE	Session log	Cross-hat memory
CF queue routing	Agentic loop	One-finding assignment
@aea-mr-coordinator	Computational sensor	Producer-consumer split
PO / PM / sponsor packet	Human approval	System-level escalation

XVI. HARNESS VS. PROMPT AND CONTEXT FOR EXPERIENCE SYSTEMS

Why a dual-viewport florist cannot be prompt-engineered into honesty

TABLE XIV-B — THREE DISCIPLINES, EXPERIENCE-SHAPED

CHALLENGE	PROMPT ENG.	CONTEXT ENG.	HARNESS ENG.
False Available	Do not overclaim	Retrieve stock text	Fail-closed sensor
Lost intent	Summarize in prompt	Memory / SU object	DATE_RE + FR-001
Wrong hat edits CSS	Instruct in chat	Limit file list	UX ownership + MRC
Quality varies	Add examples	Few-shot journeys	Guards + clip probe
Runs on forever	Be concise	Limit context	One-CF + merge policy
Errors repeat	Re-add correction	Store in notes	Guide + CF ratchet
Unprobed status	Add warning	Retrieve last brief	Honesty trip wire

A prompt cannot disable Select. A retrieval pipeline cannot stop a knowledge MR from closing a UI finding. A memory system cannot fire a DATE_RE trip wire. Those are harness responsibilities. Prompt, context, and harness all exist on Path B. If forced to spend the next engineering hour, spend it on the layer that makes a lie unrepresentable.

A. Compounding

Guide rules, guards, and merge policy compound across sessions. Prompt corrections do not. Context corpora degrade as they grow. Harness improvements are structural. That is the lesson borrowed from related work, not a reprint of its bench deltas.

B. When All Three Are Needed

The concierge voice is a prompt problem. Shared Understanding is a context problem. Fail-closed inventory, MRC, and honesty are harness problems. Removing any layer weakens Path B. The next hour still belongs to the harness when the shop is already fluent and still untrustworthy.

XVII. LIMITATIONS AND COMMON MISTAKES

What goes wrong even with fourteen hats

A. The Over-Engineered Fifteenth Hat

A harness with a shadow implementer, twelve inferential judges, and a forty-step approval chain is not safety. It is latency. The harness should be the minimum that keeps Path B honest. If hats spend more time coordinating than shipping probed findings, the harness is too heavy. Do not invent a 15th implementer to just finish the CSS.

B. DATE_RE Pollution

Cadence jobs writing DATE_RE, or sessions dumping random-thoughts into the handoff filename, recreate CF-048. Prune. Expire. Keep one live brief name. A stale brief that claims Active Focus as fact when DATE_RE 2026-08-28 said Unknown is a sensor miss.

C. Claiming Verified Without a Probe

CF-054 is the object lesson. Knowledge merged. Spec merged. CSS merged 2026-08-28. Issue #272 closed. The row on main still said in-mr. Live J1 re-record is Unknown. Calling that verified would be a new honesty bug. Sensors that only confirm formatting will pass a shop that still fails J3 reorder and J4 tracking.

D. Memory Without Cleanup

random-thoughts/ accumulates. Old notes reference MRs that were superseded. A new session that treats them as DATE_RE will relitigate closed work. Keep the handoff thin. Leave archaeology in labeled logs.

E. The Harness Does Not Fix Bad Objectives

If the workbook said FR-019 is live Stripe, sensors would faithfully implement a false shop. The harness amplifies the objective. Review the

FR/NFR source of truth before ratcheting. Do not invent IDs to make the objective prettier.

TABLE XV — COMMON AEA HARNESS MISTAKES

MISTAKE	SYMPTOM	FIX
15th hat	Shadow implementer	Stay at 14
DATE_RE pollution	Handoff lies	Filename freeze + trip wire
Verified without probe	Queue vs reality	Honesty rule
No computational sensors	Expensive judges	Guards first
Casual merge	Producer bias	MRC only
Wrong FR status	Workbook drift	Human confirm + ID freeze
Clip as trophy	Stale evidence	Re-record after CSS

XVIII. AEA CASE EVIDENCE (HONEST)

Lily's Florist Path B; no borrowed benches

A. Lily's Florist, Path B

The product is an AI-native florist. Customers speak naturally. The interface evolves as understanding grows. Dual presentation: one session, desktop Adaptive Workspace, phone linear concierge. Live shop: <https://aea.artof.link>. Canonical remote: <https://gitlab.com/artof-group/adaptive-experience-architecture>. This is the case, not a coding-agent leaderboard.

B. CF-048, Daily-Brief Honesty

CF-048 is the AEA exhibit for demos that never ship. Status language outran probes. Verification: GitLab #259 / !280. The harness fix is the honesty rule plus the DATE_RE trip wire. That is a real ratchet step. It is not a 44-point bench swing, and this paper does not dress it up as one.

C. Dual-Viewport Trail (CF-054)

On 2026-08-28: knowledge !298 merged; spec !299 merged; feat(ux) CSS !300 merged; issue #272 closed. CF-054 row on main still said in-mr pointing at !298. Live shop re-record of J1 phone+desktop is Unknown. Report the trail. Do not round it up to verified.

D. Journeys and Leftovers

J1 Urgent Sam, J2 Planner Sarah, J3 Loyal Alex, J4 Tracker Chris. Honest leftovers from 2026-08-27 clips remain: J3 recall without a reorder badge; J4 ASO fail-closed not T-08; Track gated until checkout. Those leftovers are evidence of the sensor layer doing its job when humans look. They are also evidence the shop is not a finished trophy.

E. What This Case Does Not Show

It does not show GAIA deltas, Terminal Bench ranks, or a million generated lines. It does not show live Stripe, Edge SSR LCP, staff live chat, WebRTC, FR-017, or FR-012. It does not show 14/14 skills independently probed. It does not show CF-054 verified. Related work remains related work [1]–[25].

XVIII-B. Related Pattern (1 Sep 2026): Second Brain That Acts (rvaniaaaa) — Map Only

Honesty: This subsection is a vault taxonomy addendum. It does not absorb I1–I15 or K1–K10 by being newer. It does not rename AEA's fourteen hats. [[CF-054]] dual-viewport after CSS remains **Unknown**. On-prem private Kubernetes remains a **long-term goal**, not evidence in this paper.

Source (browser probe 2026-09-01): rvaniaaa (@rvaniaaaa), X article *The Second Brain That Acts. The Agent Team That Remembers*. (30 Aug 2026) and announcing post (displayed 1 Sep 2026). Six non-overlapping roles — SCOUT, ANALYST, STRATEGIST, GUARDIAN, EXECUTOR, OBSERVER — around one compiled second brain; GUARDIAN blocks irreversible acts before EXECUTOR; OBSERVER writes every outcome back so memory compounds. Stated blast radius: agents on one shared cloud computer share files and sessions.

AEA already anchors Outer Harness to Domain Services, uses fourteen stakeholder hats, MRC+required CI as independent review (K5), and Knowledge First vault memory. Useful borrowings are **flagged only**:

ID	Flag	Item
R1	adapt	Pre-irreversible gate language (Auto-review / sponsor / secret pastes) — not an LLM merge judge (K6 stays reject).
R2	adapt	Write-after / OBSERVER-equivalent after consequential acts (knowledge-guardian + session memory).
R3	adapt	Zero-overlap charter hygiene (dual-hat lesson); do not collapse to six generic names.
R4	defer	Write-zone separation doc for multi-agent shared filesystem.
R5	accept (fold)	Compile/read-before-act — already Knowledge First.
R6	accept	Refuse “no weak links” rhetoric; weak-link search is permanent.
R7	park	Public sharing of this AEA paper (sponsor/PO) — AEA still differs on Domain Services, CI sensors, MRC, Path B probes.
R8	reject	Replace fourteen hats with six generic roles.
R9	reject	Obsidian + cloud-sync folder as Path B memory plane. Paper close ≠ implement. Detail: research/random-thoughts/2026-09-01-aea-vs-rvaniaaaa-second-brain-agent-team.md and short successor research/random-thoughts/2026-09-01-aea-framework-harness-engineering.md .

XVIII-C. Related Pattern (10 Sep 2026): Dual-loop lessons, token playbook, shared wiki — Map Only

Honesty: Vault taxonomy addendum. Does *not* absorb I1–I15, K1–K10, or R1–R9 by being newer. x.com returned 403; post text recovered via fxtwitter on 2026-09-10. [[CF-054]] remains **Unknown** / **regressed**. These posts are not Path B probes and not an instruction to install a second Second Brain product.

Sources: gipp69 [22] (task loop + reviewer loop writing small reusable lesson files); beamnxw [23] (prompt-audit, cache, effort-matching; hillclimb cheaper configs against an eval); Av1dlive [24] (portable wiki with provenance, knowledge ≠ authority, two retrieval paths).

AEA already has Knowledge First vaults, DATE_RE, Cost Guardian + ADR-016, inbox vs canonical docs/, and MRC no-self-merge. Useful borrowings are **flagged only**:

ID	Flag	Item
X1	adapt	After-run lesson <i>candidate</i> in research/inbox/ ; promote only via Knowledge Guardian (#412). Not auto-edit of skills.
X2	adapt	Cost-guardian playbook: prune stale instructions, cache, match effort; eval-gated (#413).
X3	park	Paid API hillclimb (sponsor).
X4	reject	Install Agentic Stack or a second SecondBrain directory as Path B memory.
X5	adapt	Keep provenance and knowledge≠authority (already mostly true).
X6	reject	Ungoverned self-improving prompts / skill auto-merge. Paper close ≠ implement. Detail: research/random-thoughts/2026-09-10-aea-vs-x-scan-gipp-beam-avid.md and research/random-thoughts/2026-09-10-aea-framework-harness-engineering.md .

XVIII-D. Related Pattern (14 Sep 2026): Agent-OS vocab → Outer Harness — Keep Learning Adopt

Honesty: External learning from rari (@0xwhrrari), X Article *Grok Bot: The AI Team That Never Sleeps* (created 2026-09-03; status [2095497109524934750](#) / article id 209516024763977284) [25]. Cited via

Keep Learning and Apply (#434 / #441) — *not* a vendor Bot product endorsement, not an install instruction, and not Path B shop UI. Does *not* absorb I*/K*/R*/X* flags by being newer. No FR/NFR IDs invented. Public thin vocab lands in [docs/framework/{glossary,schema}.md](#) on MR !520; Pages Live status remains **Unknown** until probe after merge.

The article’s unit of work: a chatbot returns an answer; a Bot returns a completed job. Around the model it names three control layers — **Harness** (environment), **Loop** (improvement), **Graph** (coordination) — plus an **approval** boundary. AEA already owns those jobs inside the six Outer Harness layers. This revision makes the correspondence explicit and adopts five named practices:

TABLE XVIII-D1 — HARNESS / LOOP / GRAPH ↔ AEA OUTER HARNESS			
EXTERNAL CONTROL LAYER	AEA OUTER HARNESS HOME	WHAT AEA ALREADY DOES	
Harness (world the agent may enter)	Guides + Memory + Permissions + Observability	Playbooks, committed vault memory, access gates, telemetry	
Loop (when work may continue)	Loop + Sensors	Interpret → Act → Verify → Remember; 1 finding → 1 issue → 1 MR; probes before status words	
Graph (who owns what)	Guides + Permissions (+ team roles)	Stakeholder specialists, MRC ≠ implementer, fail-closed ownership	
Approval boundary	Permissions + Observability	Human/MRC gates; irreversible actions parked	

TABLE XVIII-D2 — GAP MATRIX (ALREADY / GAP / ADOPTED)			
THEME	ALREADY	GAP	ADOPTED (#441)
Formula + six layers; honesty/probe; vault; KLA; fail-closed; 1 → 1 → 1	Yes	—	No rewrite
MRC ≠ implementer	Yes	Named “independent checker”	Glossary + schema
Autonomy ladder 0–4	Implicit	Named levels + promotion gates	Glossary + schema
Approval by reversibility	Fail-closed + MRC	Named rule + park list	Glossary + schema
Handoff packet	SU + vault notes	Compact multi-agent fields	Glossary + schema
Account connection vs role authority	Permissions	Plumbing ≠ permission wording	Glossary + schema
Harness/Loop/Graph map	Six layers	Explicit correspondence	Schema section
Vendor Bot claims	N/A	Endorsement tone	External-learning framing; Documented

Autonomy ladder (adopted): 0 Observe → 1 Prepare → 2 Execute-with-approval → 3 Schedule-or-trigger → 4 Coordinate-specialists. Promote only with durable prior-level evidence; demote on degradation.

Approval by reversibility (adopted): finish reversible work; park irreversible gates — send / publish / purchase / delete / production / legal. A good run completes the reversible majority, stages the exact next action, and stops at the boundary.

Handoff packet (adopted fields): from / to / objective / artifacts / decisions / constraints / open_questions / next_gate. Pass ownership, not transcripts.

Account connection ≠ role authority: connecting a tool is plumbing; a role contract is permission. Keep the decisions separate — same honesty as Domain Services deciding stock while AI only interprets.

Independent checker: builder ≠ verifier. Same spirit as MRC vs implementer and probe-before-status-words. Add a checker before promoting autonomy on the ladder.

Paper close ≠ implement a third-party Bot OS. Detail: [research/random-thoughts/2026-09-14-session-memory-log-agent-os-vocab-autonomy-ladder-441.md](#) and short successor [research/random-thoughts/2026-09-14-aea-framework-harness-engineering.md](#).

XIX. CONCLUSION

The model provides interpretation. Domain services provide validation. The outer harness determines whether Lily's Florist is a product or a demo. Related work shows that harness-only changes can move coding-agent benches [2], [6], [7]. Those benches are not this shop. AEA's claim is smaller and stricter: Adaptive Experience = Shared Understanding + Domain Services + Outer Harness. The six layers are the minimum infrastructure for an experience that must work across sessions and viewpoints without improvising IDs or availability.

Guides prevent known failures. Sensors catch new ones. The agentic loop bounds work to one CF. Memory persists in DATE_RE, not in chat. Permissions freeze fourteen hats. Observability makes honesty checkable. Autonomy expands by evidence on the ladder; irreversible actions stay parked by reversibility; handoffs pass packets, not chat dumps. The practical recommendation remains the ratchet: when Path B lies, fix the harness, not the sentence. Start from the vault that already exists. Add one guard. Keep one merge owner. Log one probe. The harness accumulates. The experience becomes inspectable.

A reliable AEA system should make this statement true: every important shop claim traces to a guide that shaped it, a sensor that verified it, a hat that was allowed to touch it, a brief that preserved it, and a log that recorded it. When that statement is false, more model calls usually increase opacity. When it is true, the harness is an engineering mechanism rather than a personality.

"The engineers who thrive in 2026 are not the ones who write the most code. They are the ones who build the best environments for agents to write code in." On Path B the line restates: the team that thrives is the one that builds the best environment for an experience to stay honest.

APPENDIX: GLOSSARY

TABLE XIV — KEY TERMS

TERM	OPERATIONAL DEFINITION
Adaptive Experience	Shared Understanding + Domain Services + Outer Harness
Outer harness	14 hats, vault, CF loop, CI guards, DATE_RE, honesty
Inner runtime	Edge BFF + domain services + PostgreSQL + broker (ADRs)
Guide	Feedforward: AGENTS.md, rules, skills, ADRs, Path B spec
Sensor	Feedback: python guards, CI, fail-closed, clip probes
CF loop	One finding → one issue → one branch → one MR
DATE_RE	research/daily-briefs/YYYY-MM-DD.md; only live handoff filename
Honesty rule	Status words need a probe (CF-048 class)
Fail-closed	Missing/stale observed_at → unknown → Select disabled
MRC	@aea-mr-coordinator; independent merge verifier
Path B	Dual viewport: desktop Adaptive Workspace + phone concierge
Autonomy ladder	Levels 0–4: Observe / Prepare / Execute-with-approval / Schedule-or-trigger / Coordinate-specialists; promote by evidence
Approval by reversibility	Finish reversible work; park send/publish/purchase/delete/production/legal
Handoff packet	Compact from/to/objective/artifacts/decisions/constraints/open_questions/next_gate
Account≠authority	Connection is plumbing; role contract is permission
Independent checker	Builder ≠ verifier (MRC spirit)
Unknown	Required label when a probe was not run

REFERENCES

- [1] M. Hashimoto, "My AI Adoption Journey," mitchellh.com, Feb. 5, 2026. Related work.
- [2] R. Lopopolo, "Harness Engineering: Leveraging Codex in an Agent-First World," OpenAI, Feb. 11, 2026. Related work; 1M-line figure is not an AEA result.
- [3] B. Böckeler, "Harness Engineering for Coding Agent Users," martinofweller.com, Apr. 2026.
- [4] E. Mollick, "A Guide to Which AI to Use in the Agentic Era," 2026.
- [5] V. Trivedy, "The Anatomy of an Agent Harness," LangChain Blog, Mar. 10, 2026.
- [6] A. Masood, "Agentic Harness Engineering," Google Cloud / Medium, Jun. 2026. GAIA deltas are not AEA results.
- [7] LangChain Eng., "Improving Deep Agents with Harness Engineering," Feb. 17, 2026. Terminal Bench ranks are not AEA results.
- [8] Faros AI, "AI Engineering Report 2026: Acceleration Whiplash," Aug. 2026.
- [9] Can.ac, "I Improved 15 LLMs at Coding in One Afternoon. Only the Harness Changed," 2026.
- [10] L. Tan, "How Cursor Turned AI Agents Into Better Engineers," Aug. 12, 2026.
- [11] Google Cloud, "Harness Eng. for Multi-Agent Systems Using Google ADK 2.0," Jul. 2026. Not an official Google research paper; cited as related public material.
- [12] Anthropic, "Building Effective AI Agents," anthropic.com, Dec. 2024.
- [13] A. Osmani, "Agent Harness Engineering," addyosmani.com, Apr. 2026.
- [14] OpenAI, "Codex Agent Best Practices," 2026.
- [15] Independent compilation, "Production Agent Engineering Practice 2026 / Harness Engineering / Agent = Model + Harness: The 6-Layer Production Playbook" (harness_final.pdf). Independently compiled, not affiliated with Google, OpenAI, or Anthropic. Used as document-design template only.
- [16] Art of Group, Adaptive Experience Architecture, GitLab: <https://gitlab.com/artof-group/adaptive-experience-architecture> (AGENTS.md; .cursor/rules; .cursor/skills/aea-*/; research/coherence-findings-loop.md; research/daily-briefs; docs/01-product-vision/product-vision.md; docs/05-ux-design-guide/path-b-dual-viewport-specification.md; archive/Quantic_Project_Consolidated_Coherence_Validated.xlsx).
- [17] Lily's Florist Path B live shop, <https://aea.artof.link>, and Grafana, <https://aea.artof.link/grafana/>.
- [18] GitLab issues/MRs cited as already probed: CF-048 #259/1280; dual-viewport 1298/1299/1300 and #272; thin CRM #254/1297.
- [19] 0xWast3 (wast3), "Memory Engineering for Kimi: Why a 1M-Token Window Isn't Memory," X article, Aug. 13, 2026.
- [20] GitLab Backlog evaluations #288–#292 (Memory Engineering).
- [21] Kocer (@kocer_eth), "Five Layers of Agent Engineering: Each One Wraps the One Below It," X post / article, Aug. 30, 2026, https://x.com/kocer_eth/status/2094053231949177111.
- [22] gipp69 (@gipp69), X post 2097659069456654812, retrieved Sep. 10, 2026 (x.com 403; text via fxtwitter). Dual-loop after-run lessons. Not AEA evidence.
- [23] beamnxw (@beamnxw), X post 2097746415632155110, retrieved Sep. 10, 2026. Cost playbook + agent-memory commentary. Paid hillclimb parked. Not AEA evidence.
- [24] Av1dive (@Av1dive), X post 2097639365644279857, retrieved Sep. 10, 2026. Shared-wiki / Agentic Stack prompt. Not an install instruction.
- [25] rari (@0xwhrrari), "Grok Bot: The AI Team That Never Sleeps," X Article, Sep. 3, 2026, <https://x.com/0xwhrrari/status/2095497109524934750> (article id 2095160247639777284). External learning via Keep Learning and Apply (#441); not a vendor endorsement or AEA Path B probe.

Source Method. This document is an independent practical synthesis of the AEA vault facts listed in the task brief and of the public harness-engineering materials cited above. It is not affiliated with Google, OpenAI, Anthropic, or HashiCorp, and is not endorsed by them. Product behavior may change. 14 Sep 2026 regeneration on Grok box from dated 10 Sep HTML baseline + #441 Keep Learning adopt; CloudAgent HELD. Recommendations, templates, pseudocode, and decision rules are the author's adaptation for AEA Path B and are not official specifications of any organization mentioned in related work. All diagrams are original. Unknowns are labeled.